THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS FULL SYLLABUS PAPER SET 06

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The 12th triangular number is: 1

(a) 78 **(b)** 76 **(c)** 90 **(d)** 79

2. The next number in the powers of 3 sequence $1, 3, 9, 27, 81, \ldots$ is: 1

(a) 252 **(b)** 246 **(c)** 234 **(d)** 243

3. The hour hand of a clock turns through how many degrees in one hour? 1

(a) 30° **(b)** 60° **(c)** 15° **(d)** 45°

4. An angle of 200° is: 1

(a) reflex **(b)** acute **(c)** right **(d)** obtuse

5. A supercell is a: 1

(a) cell with a prime
(b) cell in a grid that is greater than its neighbours
(c) cell with 0
(d) cell with a palindrome

6. The number of days in a leap year is: 1

(a) 366 **(b)** 365 **(c)** 364 **(d)** 367

7. In a pictograph where one symbol stands for 5, how many symbols are needed for 35? 1

(a) 8 symbols **(b)** 9 symbols **(c)** 6 symbols **(d)** 7 symbols

8. If a bar in a bar graph reaches 30 on a scale of 1 cm = 10, its height is: 1

(a) 10 cm **(b)** 0.3 cm **(c)** 3 cm **(d)** 30 cm

9. A number is divisible by 3 if: 1

(a) its last digit is 3 **(b)** its last two digits are 33
(c) the sum of its digits is divisible by 3 **(d)** it is even

10. The smallest composite number is: 1

(a) 6 **(b)** 3 **(c)** 4 **(d)** 2

11. Two rectangles of the same perimeter: 1

(a) cannot exist **(b)** have equal diagonals
(c) always have equal areas **(d)** may have different areas

12. A rectangular field is 50 m long and 30 m wide. The length of fencing for its boundary is: 1

(a) 80 m **(b)** 110 m **(c)** 1500 m **(d)** 160 m

13. A fraction cannot have denominator: 1

(a) 10 **(b)** 0 **(c)** 1 **(d)** 2

14. Which fraction is equivalent to $\frac{2}{3}$? 1

(a) $\frac{2}{9}$ **(b)** $\frac{4}{9}$ **(c)** $\frac{6}{9}$ **(d)** $\frac{6}{12}$

15. Each angle of a rectangle measures: 1

(a) 45° **(b)** 60° **(c)** 120° **(d)** 90°

16. If one side of a square is 5 cm, then the length of each of the other sides is: 1

(a) 5 cm **(b)** 25 cm **(c)** 10 cm **(d)** 2.5 cm

17. Which letter has both horizontal and vertical lines of symmetry? 1

(a) A **(b)** E **(c)** F **(d)** H

18. The point about which a figure is rotated is the: 1

(a) vertex of reflection **(b)** radius

(c) centre of rotation **(d)** arm

19. The integer 8 steps to the left of 3 on the number line is: 1

(a) 5 **(b)** 11 **(c)** −5 **(d)** −11

20. $(-3) - (-5)$ equals: 1

(a) −2 **(b)** 8 **(c)** −8 **(d)** 2

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. How many lines are there in a complete graph of 6 points? 2

22. What is the digit sum of 98,765? 2

23. A bar of height 6 cm uses a scale of 1 cm = 5 books. How many books? 2

24. Find the smallest number divisible by both 6 and 8. 2

25. How many 1 cm squares fit into a 6 cm by 4 cm rectangle? 2

26. Locate $\frac{3}{4}$ on the number line. 2

27. Define an arc. 2

28. Find the integer lying midway between −6 and 2. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Observe 1×1=1, 11×11=121, 111×111=12321. Predict 1111×1111. 3

30. Explain how to make a rotating-arm angle model using two strips and a pin. 3

31. The temperatures were 30°, 32°, 29°, 35°, 33°. Find the highest, lowest and range. 3

32. A wire 48 cm long is bent into a square. Find its side and area. Then it is bent into a rectangle 14 cm long. Find the area. 3

33. A rectangle has length 15 cm and perimeter 50 cm. Find its breadth and say whether it is a square. 3

34. Draw the shape of a pentagon with all its lines of symmetry. 3

35. Express 20 as the sum of two primes in two different ways. 3

OR

Write the prime factorisation of 100 using a factor tree. 3

36. Ravi walks 8 steps east (+) then 11 steps west (−). Where is he from the start? 3

OR

State the signs of the sum of (−12) and 7. Explain. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–100 words each.

37. **Case study:** A compass shows directions N, E, S, W. 5

(i) What is the angle between N and S?
(ii) between N and E?
(iii) Turn 135° clockwise from N: which direction lies between?
(iv) How many right angles make a full turn?

OR

(i) Define an angle. 5
(ii) Name its parts.
(iii) Draw ∠ABC of 65° using a protractor.

38. (i) Describe the 1089 trick. 5
(ii) Check it for 721.
(iii) Explain what goes wrong if the digits are all the same.

OR

(i) Explain the Kaprekar routine. 5
(ii) Apply it to 2,531.
(iii) Show that it reaches 6174.

39. **Case study:** A recipe needs $\frac{3}{4}$ cup of flour and $\frac{1}{2}$ cup of sugar. 5

(i) How much dry material in total?
(ii) How much more flour than sugar?
(iii) If the recipe is doubled, how much flour?
(iv) Write the answer to
(i) as a mixed fraction.

OR

(i) What are equivalent fractions? 5
(ii) Find four fractions equal to $\frac{3}{4}$.
(iii) Simplify $\frac{45}{60}$.

40. (i) Explain reflection with the mirror-line property. 5
(ii) Draw the reflection of triangle ABC about a vertical line.
(iii) Verify equal distances.

OR

Case study: The Ashoka Chakra has 24 spokes. 5

(i) Order of rotational symmetry?
(ii) Angle of rotation?
(iii) Lines of symmetry?
(iv) Why is the wheel considered symmetrical?

END OF QUESTION PAPER Total: 80 Marks

MASTERMIND